

JASMEHAR KAUR

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Education

University of Waterloo

Honours Software Engineering

Sep. 2024 – Present

Waterloo, ON

Technical Skills

Languages: Python, Java, Go, C++, C, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks/Libraries: PyTorch, Scikit-learn, LangChain, Flask, FastAPI, Apache Kafka, Angular, Docker, Kubernetes, Helm, Node.js, React.js, React Native, TailwindCSS, Tesseract OCR

Tools/Technologies: Git, GitHub, Bash, MATLAB, Postman, AWS (Athena, Lambda, API Gateway), WebSocket, Raspberry Pi, Linux, REST APIs, Agile, RAG, OpenSearch, Selenium, Snyk, Aqua, DAST, SAST

Experience

Royal Bank of Canada

January 2026 – May 2026

Software Developer

Toronto, ON

- Built a **Kafka producer** in **Java** to stream household portfolio data for **4 million** clients to downstream teams, enabling real-time cross-team data access at scale.
- Designed and implemented backend features integrating with **Microsoft SQL Server**, supporting relational workflows processing **50 thousand records** per transaction cycle.
- Optimized **SQL** queries, joins, and stored procedures to reduce query latency by **37%**, resolving **multi-threaded** service bottlenecks across normalized schemas.
- Resolved **30+ dependency CVEs** surfaced by **Aqua** and **Snyk**, and built **DAST** and **SAST** test pipelines from scratch, reducing security exposure across the portfolio modelling application.

Nokia

Apr. 2025 – Aug. 2025

AI/ML and Full-Stack Developer

Ottawa, ON

- Built a cloud-native **multi-agent LLM**-based system to automate 5G alarm resolution, with **vector-based RAG**-driven chain-of-thought analysis for high-relevance retrieval.
- Engineered a **RAG** pipeline by optimizing **hybrid search** (dense + sparse) in **OpenSearch** and implementing multi-pass retrieval, achieving a **65%** increase in precision and **58%** increase in recall.
- Developed a **Crossplane-based cloud automation tool** and a **custom Kubernetes operator** in **Go** to manage the lifecycle of autonomous applications and enable intent-based network orchestration.
- Built interactive **React** frontends with modular components and dynamic theming.

Trexo Robotics (YC '19)

Jul. 2022 – Sep. 2022

Software Developer

Mississauga, ON

- Developed a **serverless data management system** with React, Redux, and **AWS Athena** via **API Gateway**, enabling authenticated **CRUD operations** on large datasets and cutting data editing time by **92%**.
- Built **custom API endpoints** and scripts to automate manual data entry, establishing a **controlled, programmatic interface** for database interactions and reducing data entry errors.

Absolute Robotics

Sep. 2022 – Jul. 2024

IT Lead

Mississauga, ON

- Engineered a **data collection and analytics** system with a **React Native** app for real-time match data capture and a **React** web app for offline ingestion via QR codes, reducing data entry time by **95%**.

Projects

Pelican | Python, LangGraph, Polars, DuckDB, CVXPy, FastAPI, React

- Designed a multi-agent pipeline (Researcher → Coder → Critic) that mines arXiv, gates signal acceptance on IC t-stat ≥ 1.5 and net Sharpe ≥ 0.3 , and retries with a distinct economic mechanism on rejection; evaluated **56 hypotheses** across **10 registered signals**.
- Built a vectorized **Polars** cross-sectional backtest engine over **700 tickers / 36 periods** with strict **point-in-time** joins (fundamentals, SEC EDGAR tone, news sentiment) completing each backtest in **<2s** with no look-ahead bias.
- Implemented **Almgren-Chriss** execution cost modeling alongside a **CVXPy** mean-variance optimizer with **Ledoit-Wolf** covariance shrinkage and walk-forward IC-weighted allocation to penalize high-turnover strategies.

dill.pkl | FastAPI, React, TypeScript, Python, Optuna, OpenRouter

- Built an **8-agent AutoML** web app automating the full ML pipeline — data analysis, preprocessing, feature engineering, model selection, training, and evaluation — outputting a fully deployed, documented model.
- Designed a **dependency-aware rerun engine** and **controlled action registry** enabling natural-language pipeline modification, keeping LLM intent separate from execution for safe, deterministic changes.
- Parallelized training across candidate models with per-candidate **Optuna HPO**, packaging the best by CV score into a deployment zip with a **schema-validated FastAPI endpoint** and auto-generated README.

Braillinator | Python, PyTorch, U-Net, TrOCR, Tesseract OCR, Raspberry Pi, React Native

- Built and trained a conditional **U-Net** image enhancement model with **FiLM conditioning** and spatial OCR **confidence maps** to improve Tesseract accuracy on low-quality phone camera images, achieving **54%** reduction in character error rate on degraded inputs.
- Engineered an end-to-end **image to Braille** pipeline: React Native app → WebSocket → Raspberry Pi OCR → Braille tablet with piston-driven dots for **real-time** tactile output.

Certifications

AWS Cloud Technical Essentials